

# NetSage AI – Cisco Packet Tracer Troubleshooting Platform

## Official Project Summary & Individual Contribution Document

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|------------------------|--------------------------------------------|
| Candidate Name:        | Lucky Chauhan                              |
| College / Institution: | KCC Institute Of Technology And Management |
| Technology Domain:     | NetSage AI (Networking & Applied AI)       |
| Submission Category:   | Cisco Networking & Packet Tracer Capstone  |

### 1. Executive Summary & Project Objectives

NetSage AI is an advanced, production-grade autonomous network troubleshooting assistant designed specifically to assist network engineers and students in debugging Cisco Packet Tracer lab topologies. By orchestrating a two-stage diagnostic pipeline—combining a deterministic 14-rule regex evaluation engine with the Google Gemini 3.6 Flash LLM—NetSage AI provides instant, evidence-grounded root-cause analysis, OSI layer classification, and copy-ready Cisco IOS CLI remediation commands. Responsible AI is enforced via a mandatory Human-in-the-Loop review board.

### 2. Packet Tracer Topology & IP Addressing Plan

The companion Cisco Packet Tracer (.pkt) topology simulates an enterprise network with Dual Cisco 2911 Routers (OSPF Area 0, RoAS, NAT Overload), Catalyst 2960 Switches (VLANs 10, 20, 30, 802.1Q Trunks, Port Security), and Centralized DHCP/DNS/Web Servers.

| Device           | Interface   | IP Address     | Subnet Mask     | Role / Notes    |
|------------------|-------------|----------------|-----------------|-----------------|
| Router0 (HQ)     | Gig0/0.10   | 192.168.10.1   | 255.255.255.0   | VLAN 10 Gateway |
| Router0 (HQ)     | Gig0/0.20   | 192.168.20.1   | 255.255.255.0   | VLAN 20 Gateway |
| Router0 (HQ)     | Serial0/0/0 | 10.0.0.1       | 255.255.255.252 | WAN to Branch   |
| Router1 (Branch) | Serial0/0/0 | 10.0.0.2       | 255.255.255.252 | WAN to HQ       |
| Server0          | FastEth0    | 192.168.10.254 | 255.255.255.0   | DHCP/DNS/HTTP   |

### 3. Protocols & Diagnostic Vectors Handled

- **Layer 1 (Physical):** Administratively down interfaces, interface speed/duplex mismatches, cable faults.
- **Layer 2 (Data Link):** 802.1Q trunk mode conflicts, missing access VLANs, Port Security err-disabled shutdown, STP loops.
- **Layer 3 (Network):** Subnet mask errors, gateway mismatches, missing default/static routes, OSPF Area mismatches, NAT inside/outside.
- **Layer 4 & 7:** ACL filtering direction & port blocks, DHCP pool exhaustion & relay helper-addresses, DNS server resolution.

### 4. Individual Student Contribution Breakdown

**Student Name:** Lucky Chauhan | **Assigned Role:** Lead Full-Stack Architect & AI Systems Engineer

- **AI Diagnostic Engine Architecture:** Integrated Google Gemini 3.6 Flash LLM with structured Pydantic schema validation for OSI layer classification and command remediation.
- **Deterministic 14-Rule Matrix Engine:** Engineered regex & keyword heuristic rule engine for instant sub-millisecond detection of interface shutdown, VLAN errors, and routing mismatches.
- **Full-Stack Streamlit Platform & Theme:** Developed the multi-page responsive enterprise web interface featuring Obsidian Dark & Crisp White theme, CLI terminal simulators, and Plotly analytics.

- **Git & Version Control Lifecycle:** Configured repository structure, continuous deployment synchronization, environment secret management, and GitHub repository hosting.

## 5. Originality & Compliance Declaration

I hereby declare that this project submission for 'NetSage AI' by Lucky Chauhan from KCC Institute Of Technology And Management is an authentic, original implementation. All modular Python code, prompt engineering, rule checking mechanisms, and Cisco Packet Tracer network configurations are distinct and strictly adhere to academic integrity and anti-plagiarism guidelines.

**Candidate:** Lucky Chauhan

**Institution:** KCC Institute Of Technology And  
Management

**Signature:** \_\_\_\_\_

**Date:** September 2026